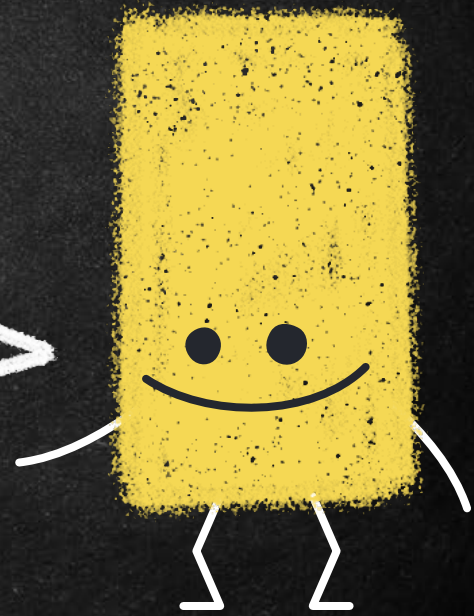


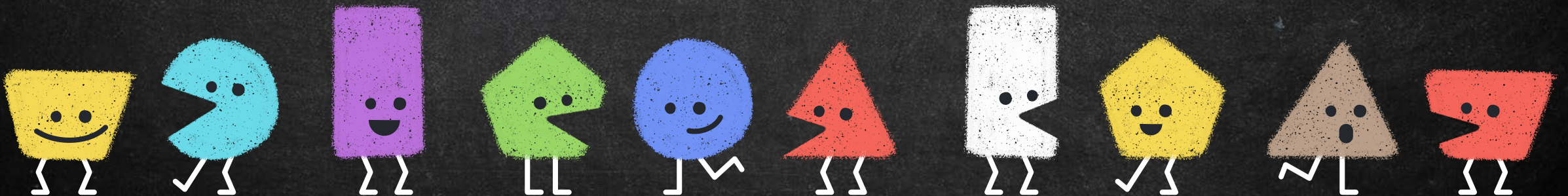
Digital Forensics Part - 1



Hello!

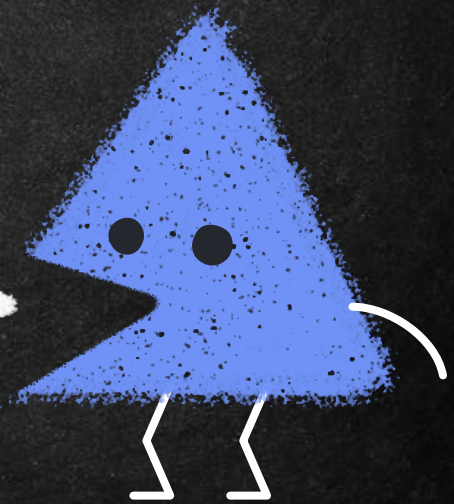
I am Sarang Rajvansh
Assistant Professor,
School of Cyber Security & Digital Forensics,
National Forensic Sciences university,
Gujarat Campus, Gandhinagar.

You can find me at sarang.rajvansh@nfsu.ac.in

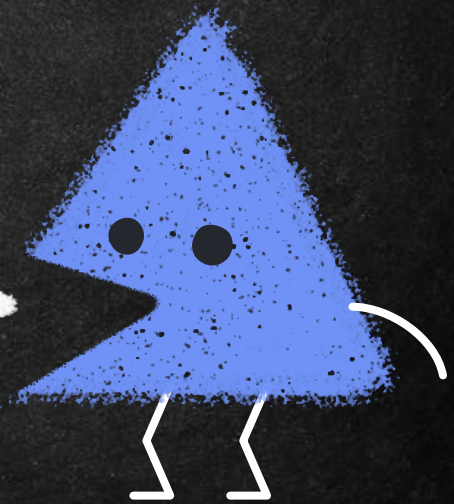
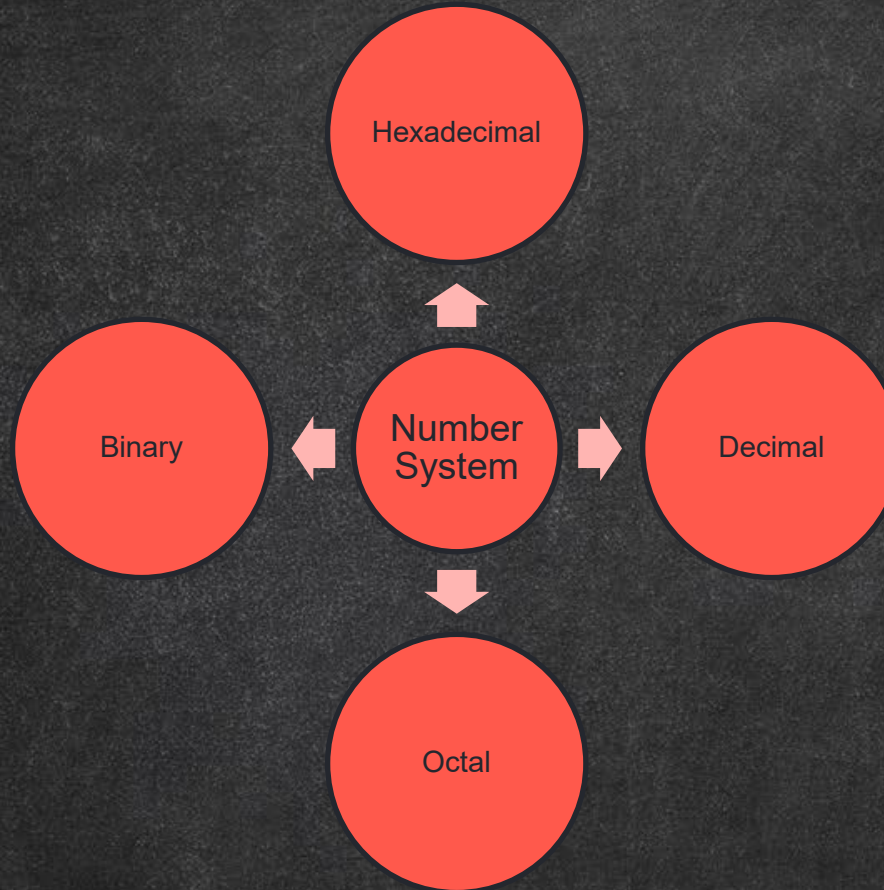


Number System

- A number system is a method to represent (write) numbers.
- Every number system has a set of unique characters or literals.
- The count of these literals is called the radix or base of the number system.
- Number systems are also called positional number system because the value of each symbol (i.e., digit and alphabet) in a number depends upon its position within the number.

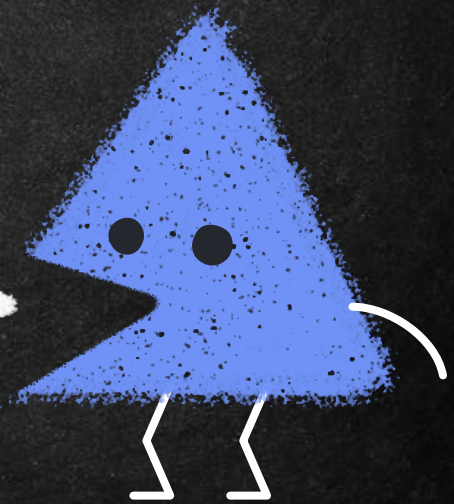


Number System



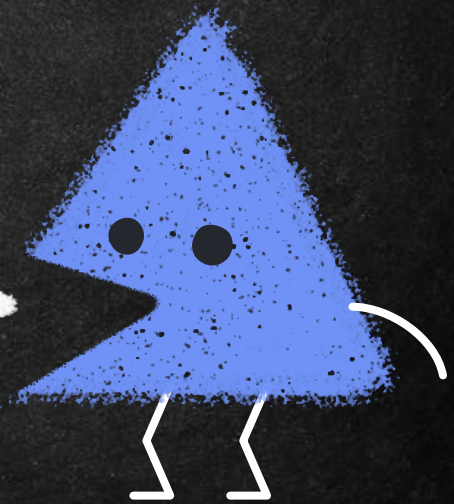
Conversion From Decimal

- Step 1: Divide the given number by the base value (b) of the number system in which it is to be converted
- Step 2: Note the remainder
- Step 3: Keep Dividing until the quotient is zero
- Step 4: Write the noted remainders in the reverse order (from bottom to top)



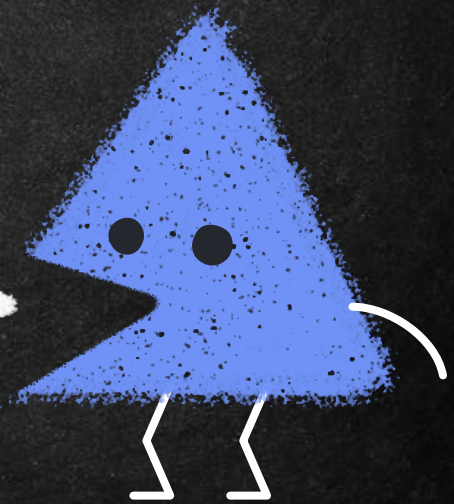
Conversion to Decimal

- Step 1: Write the position number for each alphanumeric symbol in the given number
- Step 2: Get positional value for each symbol by raising its position number to the base value b symbol in the given number
- Step 3: Multiply each digit with the respective positional value to get a decimal value
- Step 4: Add all these decimal values to get the equivalent decimal number



Task 1

- Binary to Decimal
 - 100100111001
- Hexadecimal to Decimal
 - A1B8C5
- Octal to Decimal
 - 743924



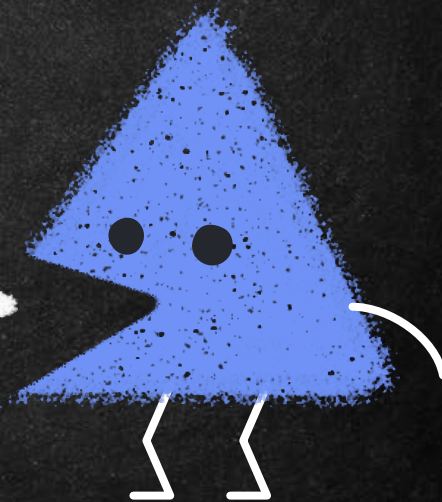
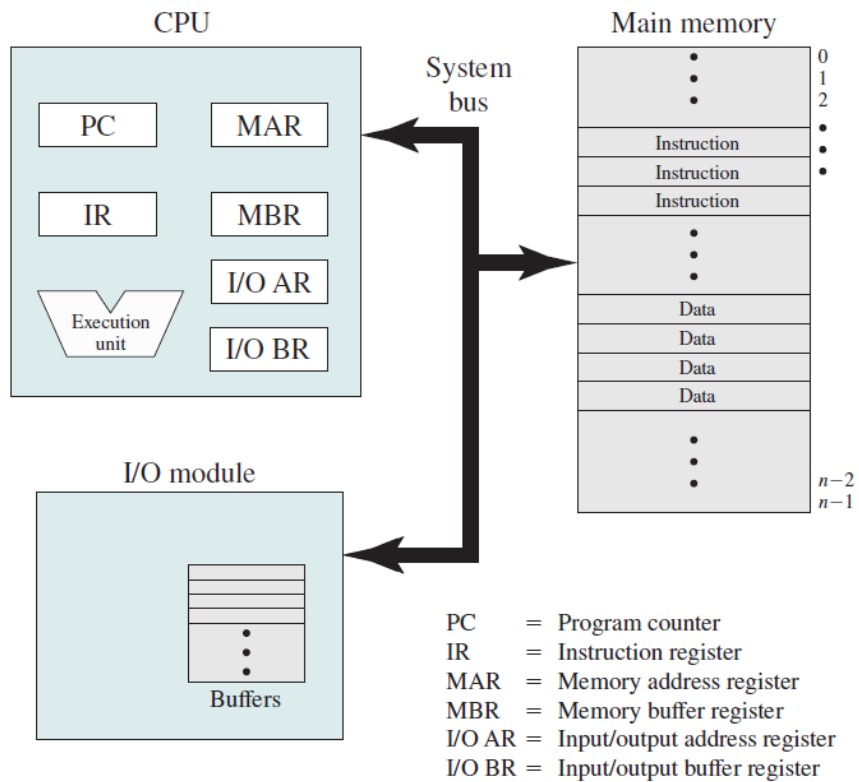
Basic Elements of Computer Systems

Processor

Main Memory

I/O Modules

System Bus



Overview: Computer Systems

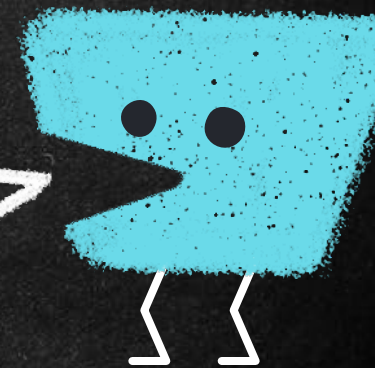
9

Processor

Controls the operation of the computer and performs its data processing functions. When there is only one processor, it is often referred to as the central processing unit (CPU).

Main memory

Stores data and programs. This memory is typically volatile; that is, when the computer is shut down, the contents of the memory are lost. In contrast, the contents of disk memory are retained even when the computer system is shut down. Main memory is also referred to as real memory or primary memory.



Overview: Computer Systems

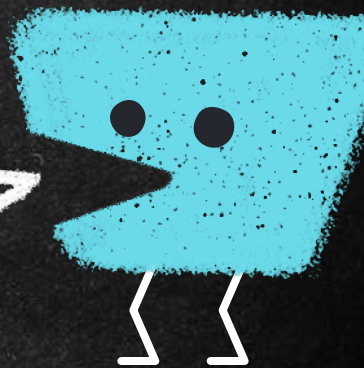
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I/O Modules

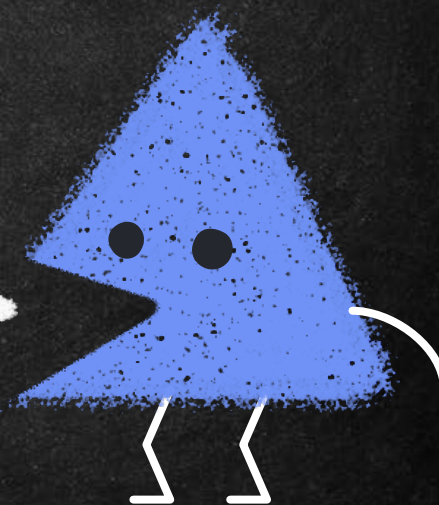
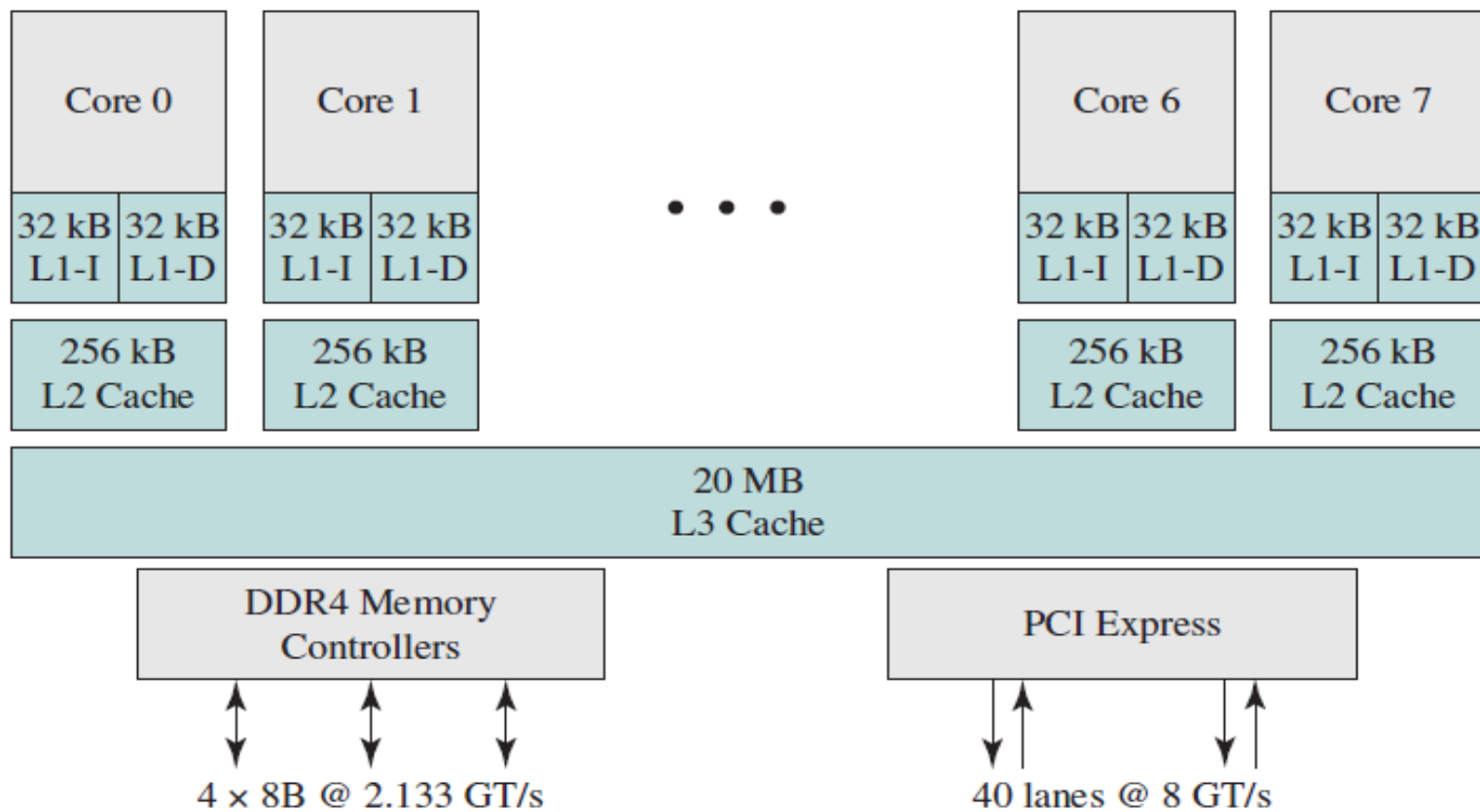
Move data between the computer and its external environment. The external environment consists of a variety of devices, including secondary memory devices (e.g., disks), communications equipment, and terminals.

System bus

Provides for communication among processors, main memory, and I/O modules.

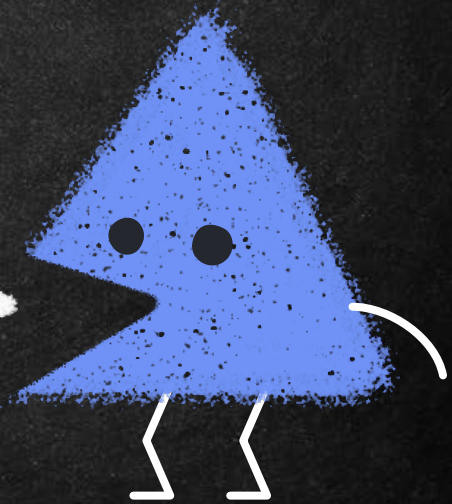


Chip Diagram of 8 Core Processor



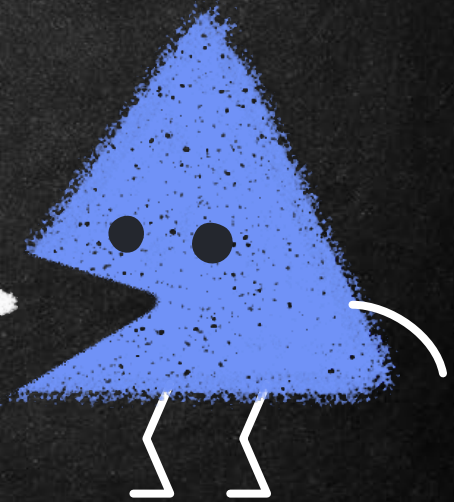
Physical and Logical Storage

- File storage and recording is largely controlled by the operating system.
- A **byte** is comprised of 8 bits and is the smallest addressable unit in memory.
- A **sector** on a magnetic hard disk represents 512 bytes, or 2048 bytes on optical discs.



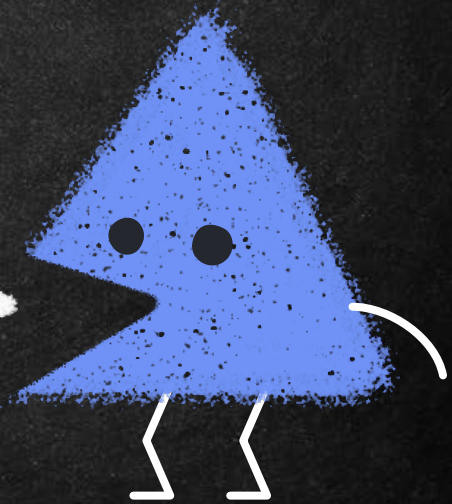
Physical and Logical Storage

- A **bad sector** is an area of the disk that can no longer be used to store data.
- Bad sectors can be caused by viruses, corrupted boot records, physical disruptions, and a host of other disk errors.



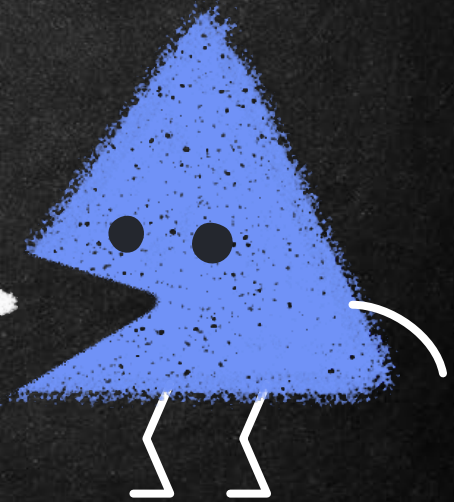
Physical and Logical Storage

- A **cluster** is a logical storage unit on a hard disk that contains contiguous sectors.
- When a disk volume is partitioned, the number of sectors in a cluster is defined.
- A cluster can contain 1 sector (512K) or even 128 sectors (65,536K).

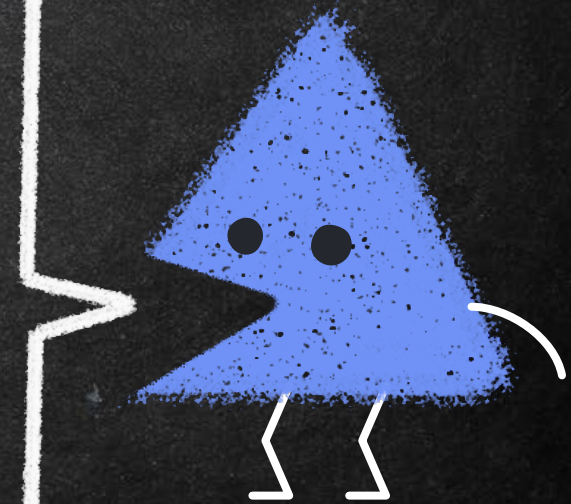
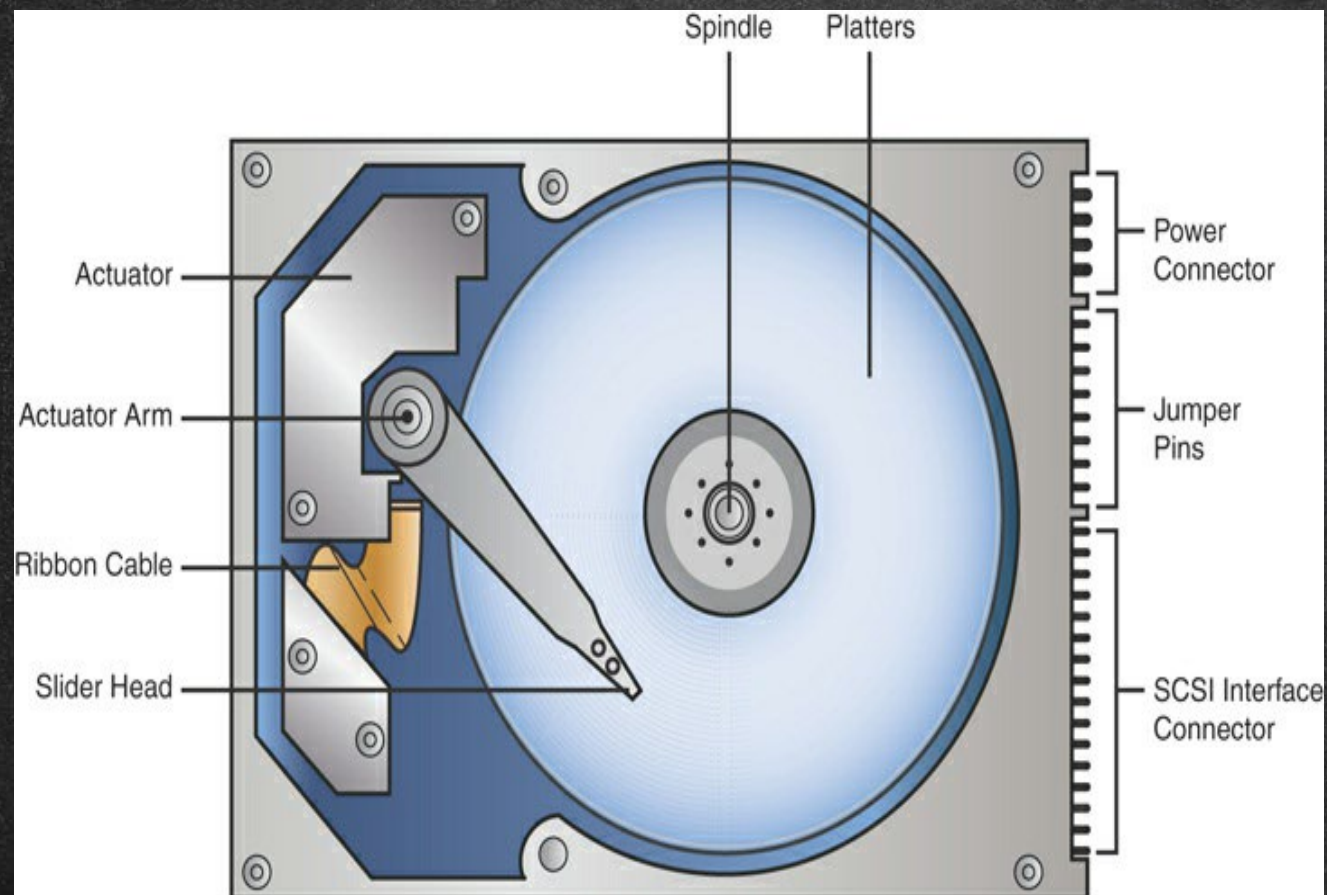


Physical and Logical Storage

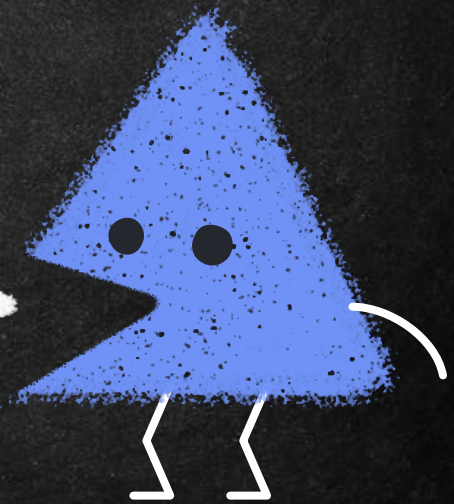
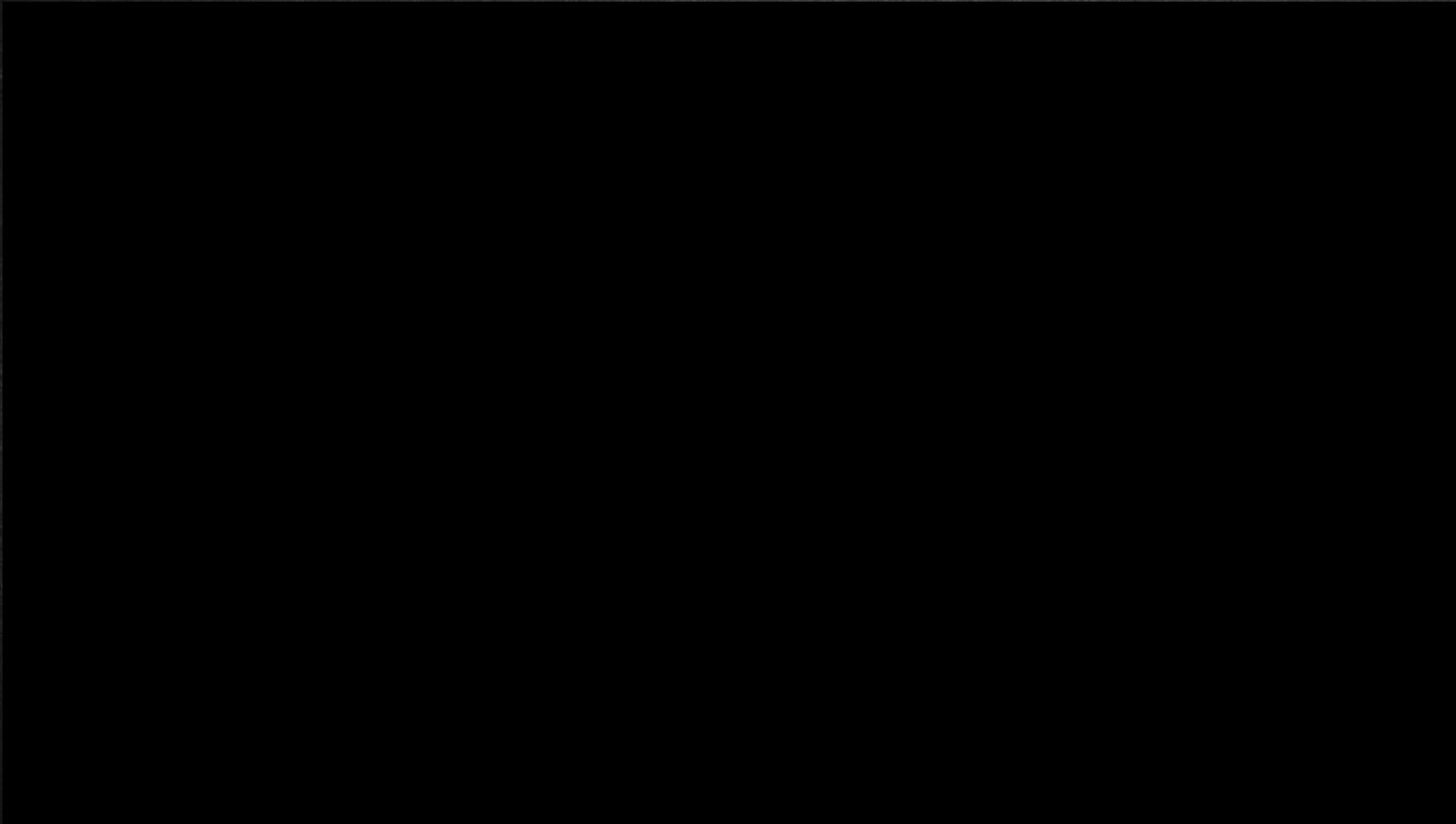
- **Tracks** are thin, concentric bands on a disk that consist of sectors where data is stored.



The physical layout of a hard disk

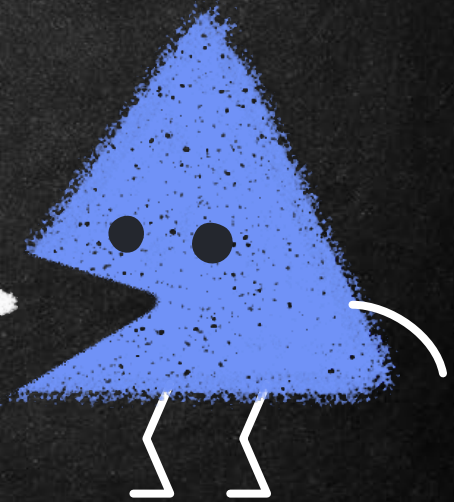


The physical layout of a hard disk



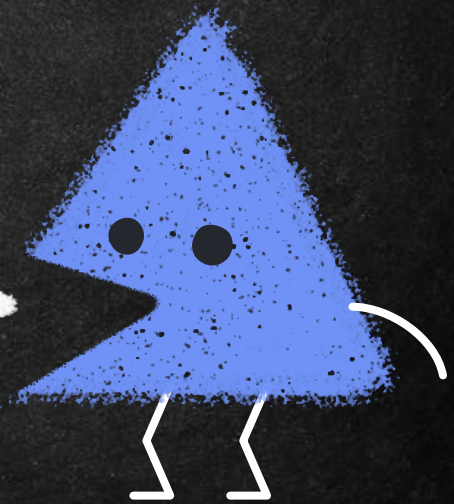
The physical layout of a hard disk

- A platter is a circular disk made from aluminum, ceramic, or glass that stores data magnetically. A hard drive contains one or more platters, and data can usually be stored on both sides of this rigid disk.



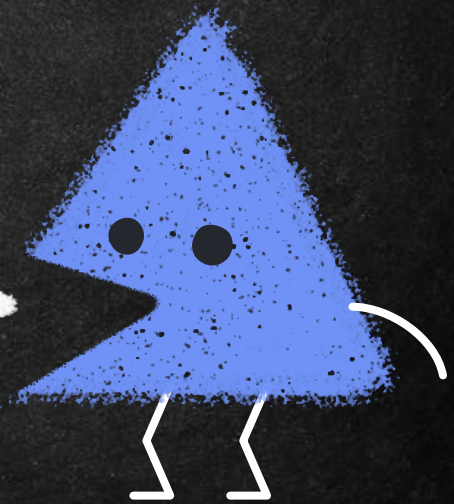
The physical layout of a hard disk

- A spindle, at the center of the disk, is powered by a motor and is used to spin the platters. An arm sweeps across the rotating platter in an arc.



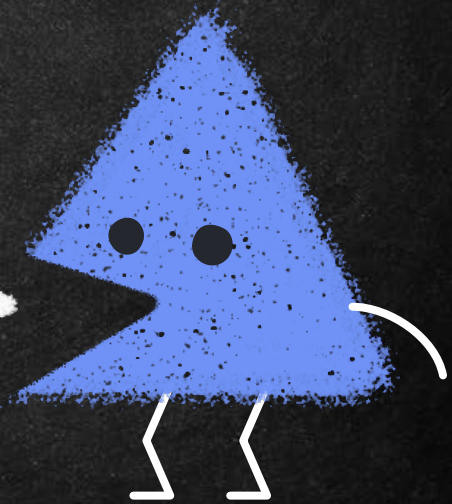
The physical layout of a hard disk

- This actuator arm contains a read/write head that modifies the magnetization of the disk when writing to it.
- There are generally two read/write heads for each platter because a platter usually contains data on both sides.



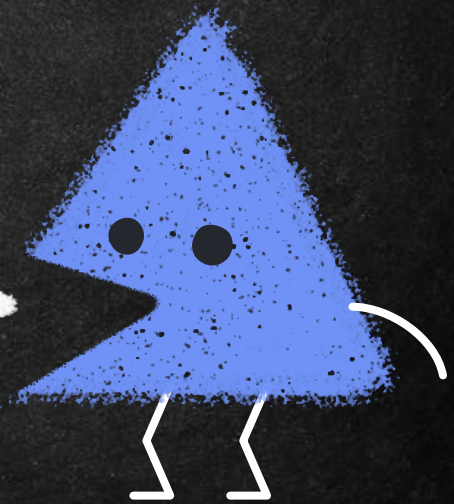
The physical layout of a hard disk

- The arm and head are nanometers from the platter. Therefore, hard drives must be handled very carefully; any impact on the hard drive could render the read/write head useless.
- Additionally, the examiner must be very careful not to let any magnetic device near a hard drive.



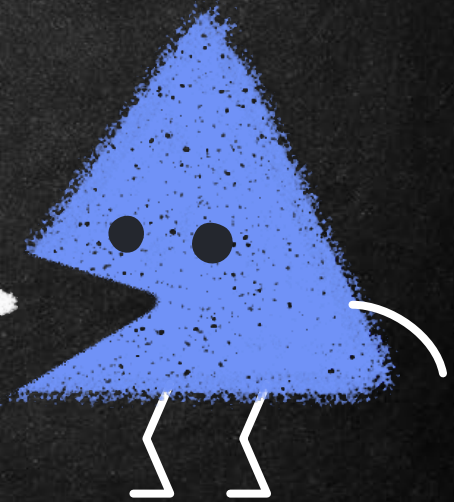
The physical layout of a hard disk

- A cylinder is the same track number on each platter, spanning all platters in a hard drive.
- Disk geometry refers to the structure of a hard disk in terms of platters, tracks, and sectors.

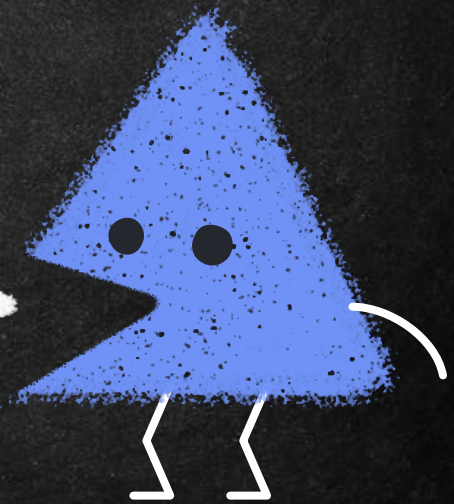
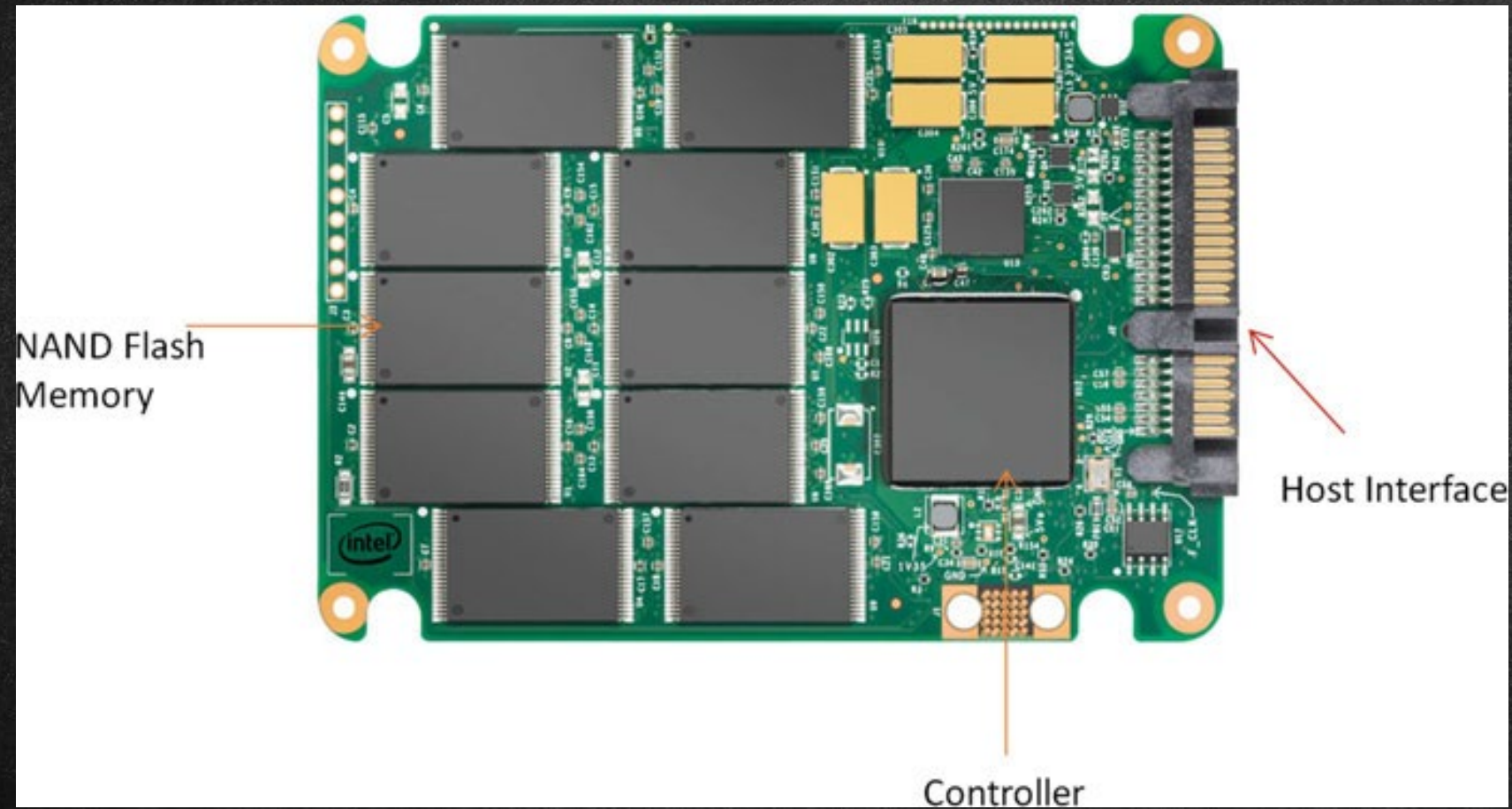


The physical layout of a hard disk

- The capacity of a hard disk drive can be calculated using the following formula:
- $\text{Number of cylinders} \times \text{Number of heads} \times \text{Number of sectors} \times \text{Number of bytes per sector}$

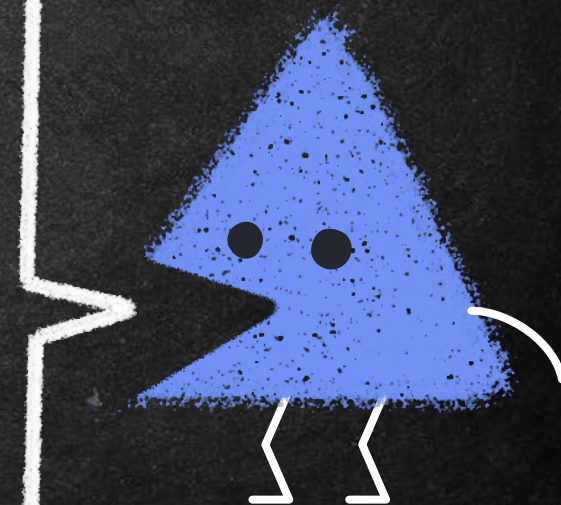


The physical layout of a Solid State Drive



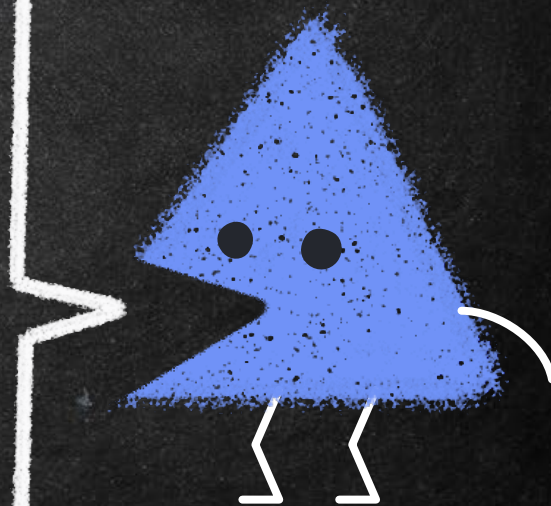
HPA Drive Areas

- *Host Protected Area (HPA)* was introduced in the ATA/ATAPI-4 standard and allowed system vendors to reserve portions of the disk for use outside the normal OS.
- For example, the HPA can be used to store persistent data, system recovery data, hibernation data, and so on.
- Access to this data is controlled by the system firmware rather than the installed OS.



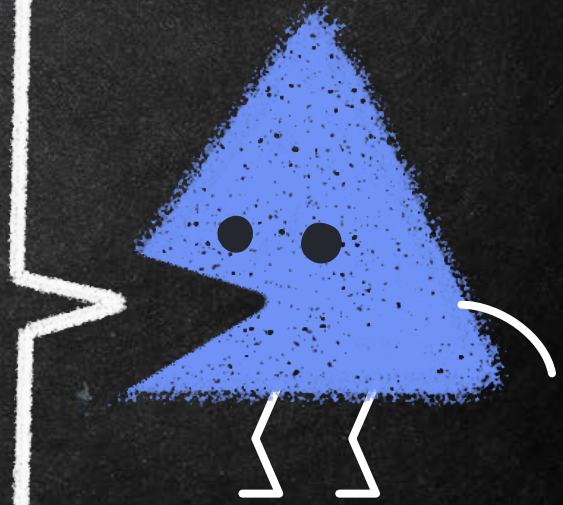
DCO Drive Areas

- *Device Configuration Overlay (DCO)* feature was introduced in the ATA/ATAPI-6 standard and provided the ability to control reported disk features and capacity.
- This allowed system vendors to ship disks from multiple manufacturers while maintaining identical numbers of user-accessible sectors and features across the disks.
- This facilitated easier support and drive replacement.
- Both the HPA and DCO can coexist; however, the DCO must be created first, followed by the HPA.
- The HPA and DCO have been misused by criminals and



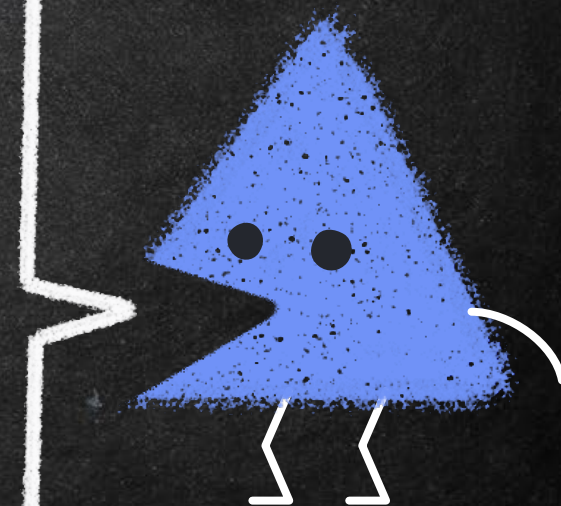
Locard's exchange principle

- Locard's exchange principle says that in the physical world, when perpetrators enter or leave a crime scene, they will leave something behind and take something with them. Examples include DNA, latent prints, hair, and fibers.
- The same holds true in digital forensics. Registry keys and log files can serve as the digital equivalent to hair and fiber. Like DNA, our ability to detect and analyze these artifacts relies heavily on the technology available at the time.
- Viewing a device or incident through the "lens" of Locard's principle can be very helpful in locating and



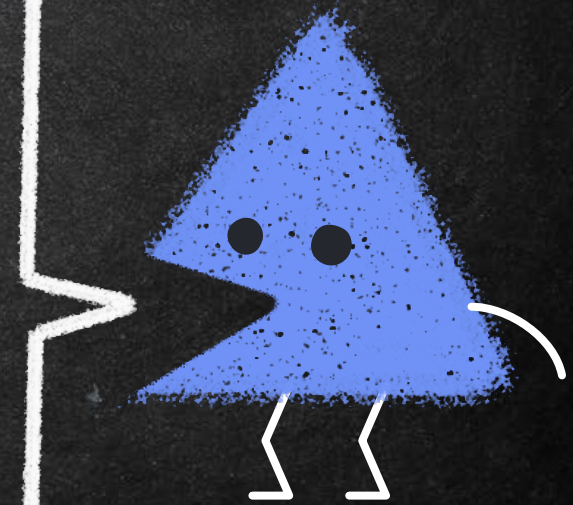
File Overview

- files will have a specific structure or format. The start of the file is called the “header.” The end of the file is known as the “footer.” All of the bytes in between represent the remainder of the file.
- A file header is considered a form of metadata (“data about the data”). The header, like the extension, is used to identify the file type. However, unlike the extension, the header is much harder to change and is generally inaccessible to most users.



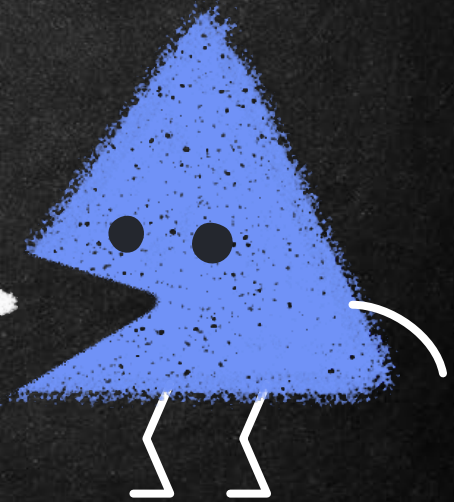
File Overview

- When the file is saved or written to the hard drive, it's not necessarily saved to contiguous clusters
- These separate pieces of the file could be on different sides of a platter or on different platters altogether. When we double click on the file to open it, the computer gets the locations of all the sectors allocated to the file from the file system and recreates your file.
- To work on the file, it must be loaded into the computer's main memory, also known as RAM. From here, the file is fed into the central processing unit (CPU) as we're working with it.



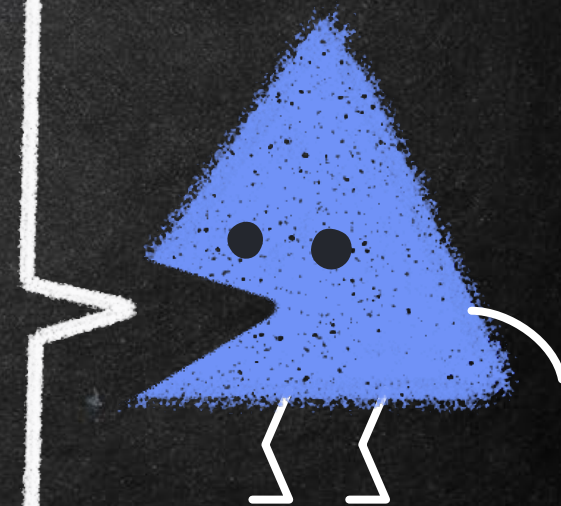
Data Type

- Data can be lumped into three broad categories: active, latent, and archival.
- Looking at data in this way helps in clarifying their location, how they're accounted for by the file system, how they can be accessed by the user, and so on.
- It also helps to narrow down the cost and effort required to recover the data in question.



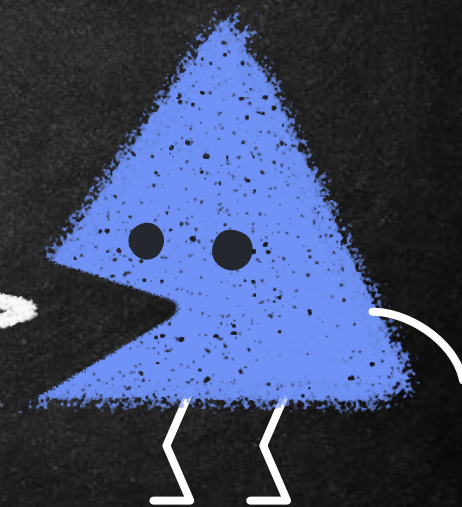
Active Data

- Active data are the data that we use every day on our computers. The operating system “sees” and tracks these files.
- These files can be located using Windows Explorer.
- These are the files that reside in the allocated space of the drive.
- These data can be acquired with standard forensic cloning techniques.



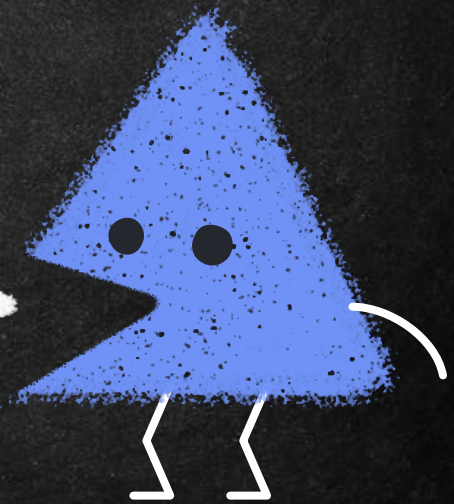
Latent Data

- Data that has been deleted or partially overwritten are classified as latent.
- These files are no longer tracked by the operating system and are therefore “invisible” to the average user.
- Looking for one of these files with Windows Explorer and it could not be found.
- A bit stream or forensic image is required to collect these data.



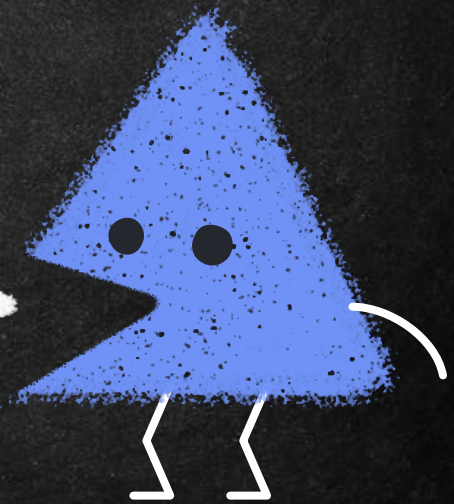
Archival Data

- Archival data, or backups, can take many forms.
- External hard drives, DVDs, and backup tapes etc.
- Acquisition of archival data can range from simple to extremely complex.
- The type and age of the backup media are major factors in determining the complexity of the process.



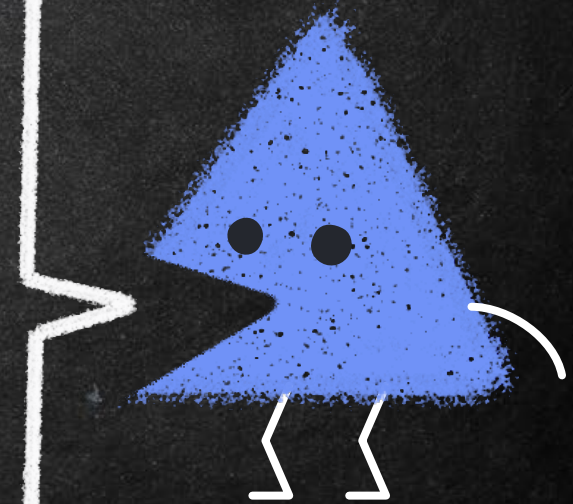
How Data is stored?

- Magnetic disk drives historically stored data in units known as cylinders, heads, and sectors. These units were then used to assign address to each storage block on a disk.
- concentric circles of storage areas on each side of the drive's platter were known as *tracks*. Tracks were limited to 63 sectors.



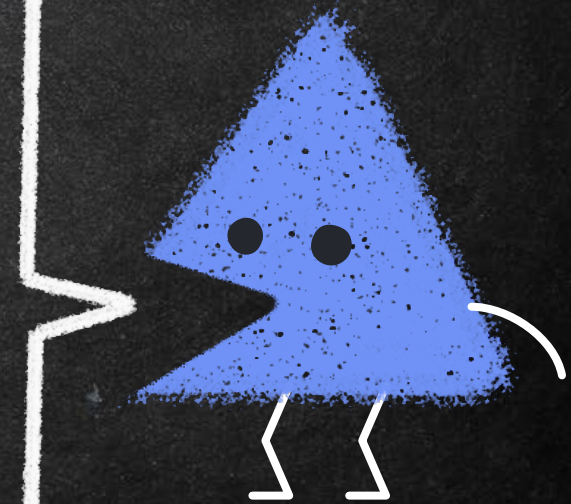
How Data is stored?

- The computer's operating system only stores data as clusters.
- **Slack Space:** The slack space is the difference between the space that is assigned and the space that is actually used.
- The hard drive is used to store your data and applications when they aren't being used.
- The sectors containing leftover data are commonly referred to as *slack space* or *cluster slack*.



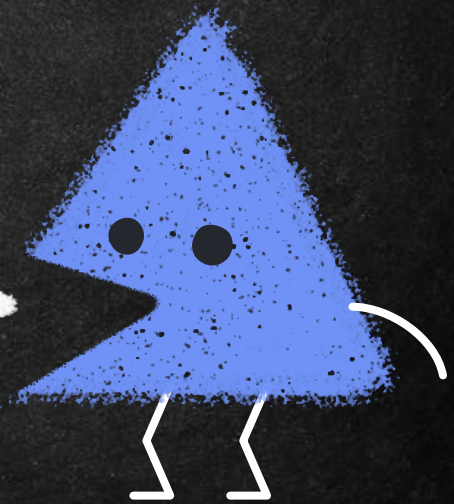
Slack Space using Example

- If we created a one-byte file, an NTFS file system would write the single byte and 511 null bytes to the cluster.
- If the cluster size were equal to one sector, it would be written in the same way. If the cluster size were 4 Kb (eight sectors), then the file system would write the first byte of the file, followed by 511 null bytes into the first sector.
- The remaining sectors in the cluster would not be written to (or overwritten if they previously contained data from a deleted file).



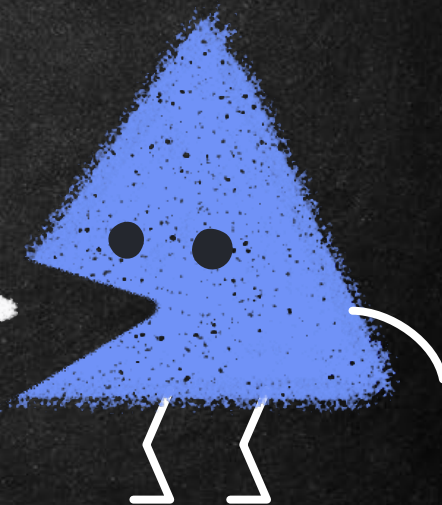
Data Carving

- Data carving is the act of searching for particular strings or bytes within a structure.
- A hex editor or other data-viewing tools can be used to carve for data.
- The analyst determines a string or binary pattern to search for, then initiates a search across a device or structure for that string or pattern.



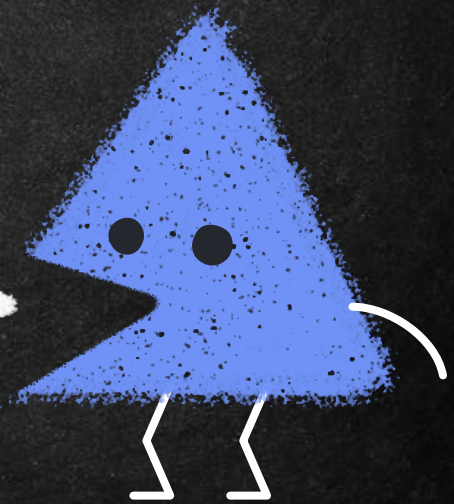
Data Carving

- The target can be of whatever scope is appropriate for the task, such as a file, slack space, unallocated space, a full volume, a memory image, or a swap file.
- The technique can be used to carve for full files – such as recovering deleted JPG image files; or for records – such as recovering portions of a deleted Windows event log.



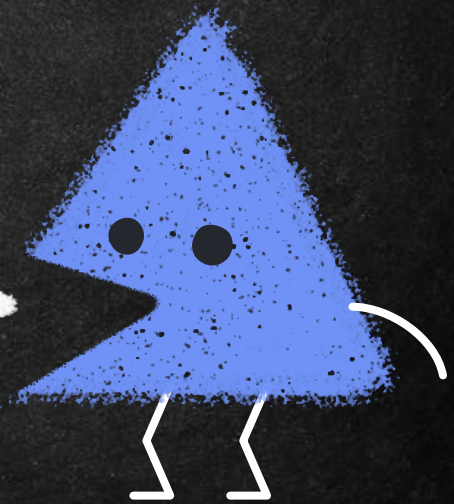
Data Carving

- Carving for records can be a very effective method of finding evidence when it's been deleted either intentionally or through normal system processes.



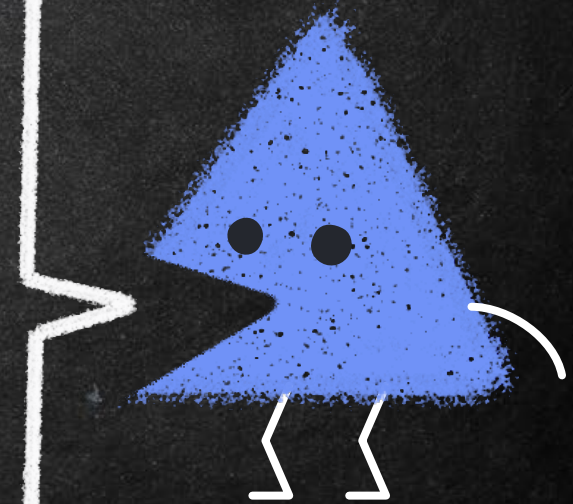
Data Carving : Files

- Full files can be carved by searching for particular file headers or signatures.
- File carving is typically limited to carving for contiguous – non fragmented – files in unallocated space or in other files.
- Techniques exist to carve data that is fragmented using fuzzy hashing



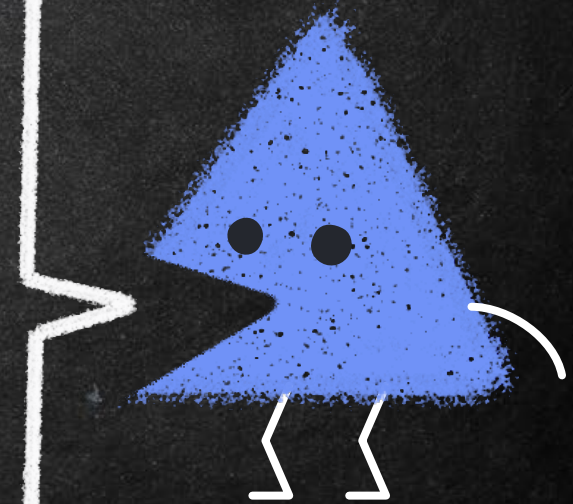
Data Carving : Records

- Carving for records involves searching for data where the full file may not be essential to recovering information.
- Like to carve for a weblog entry that contains full context in a single event: date and time, webpage visited, IP address accessing the page, browser ID, and so on.
- While a weblog is typically maintained in plain text, other data may be contained in binary blobs (such as Windows event logs) or a database format.



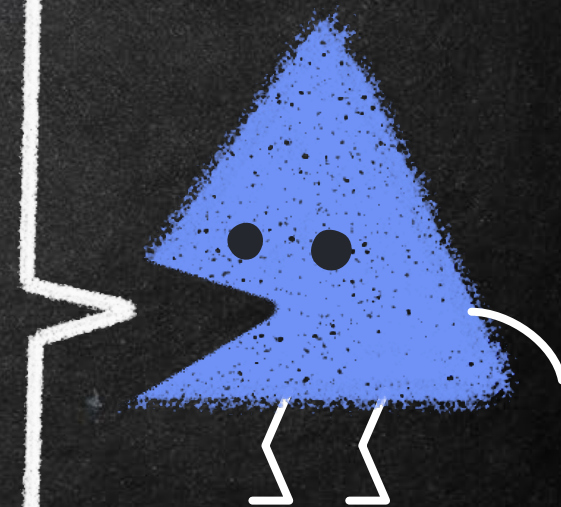
Data Carving: Example

- The forensic investigator received a CPU from law enforcement for examination.
- The system was seized because it was running a Windows System. When the system was analyzed, the system logs had been deleted and very little other logical evidence remained.
- The investigator carved for Windows Event Log records and successfully recovered over twenty-two thousand entries.



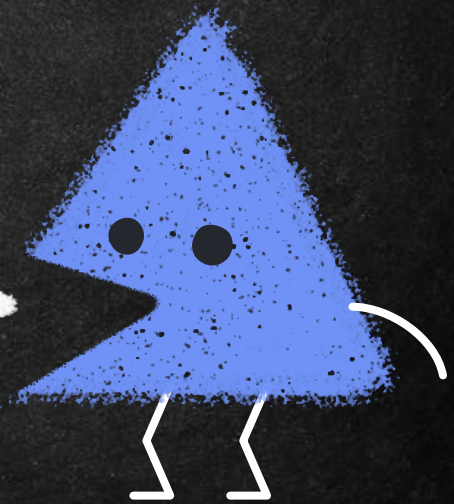
Data Carving: Example

- These entries were key in helping law enforcement determine the evidence for intrusion in the case because they contained information about all malware infections from the infected system checking into the system logs.
- Information retrieved included:
 - IP address of remote server,
 - system host name, and
 - date and time of the check-in etc.



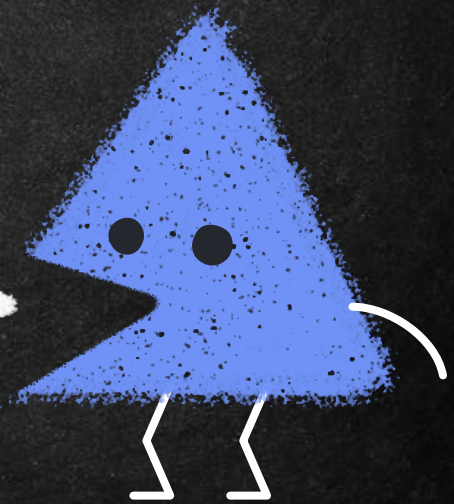
Data Acquisition

- Data acquisition is the act of collecting data relevant to an investigation.
- This data can be live data, including memory captures, or it can be collected in a forensically sound manner as an image or a forensic copy.
- The data acquisition should be conducted in a manner that creates the fewest changes to a system.



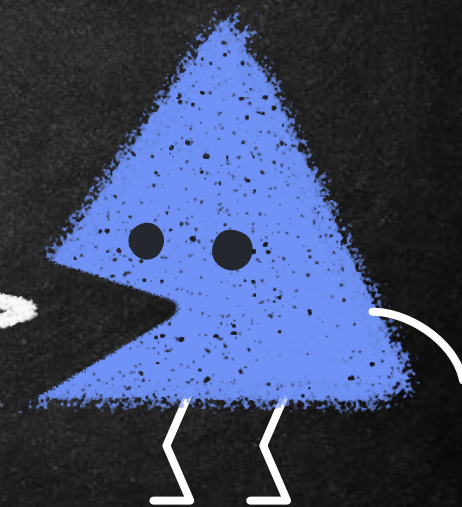
Data Acquisition

- The acquisition of a system that is powered off is the simplest example of collecting data in a forensically sound manner:
 - The analyst removes the system's hard drive.
 - The hard drive is connected to the read-only port of a firmware write block.
 - A target device to store an image or a copy is connected to the read-write port of the write block.



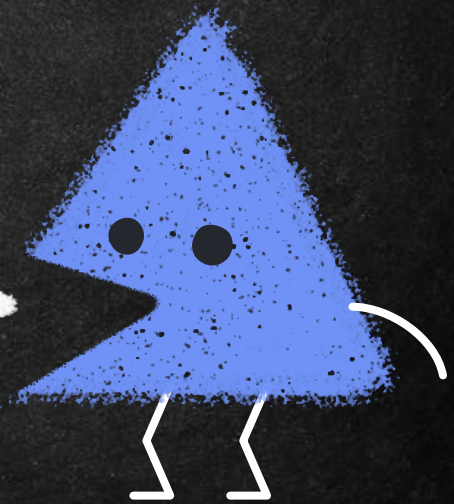
Data Acquisition

- The communications port of the write block is connected to a computer or a stand alone device, or is built in.
- The analyst then creates an image or a copy by writing to the target device.
- The original is maintained and documented in a chain-of-custody form.



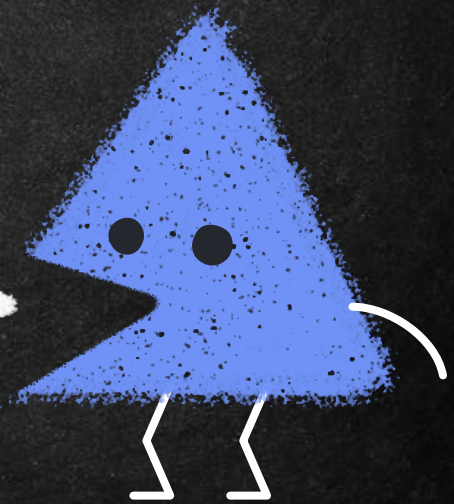
Data Acquisition

- The image or copy can then be examined (or additional copies made) on the analysis platform.
- This is typically done as a read-only operation as well, which allows for repeatability of any examination or experiments without modifying the data.



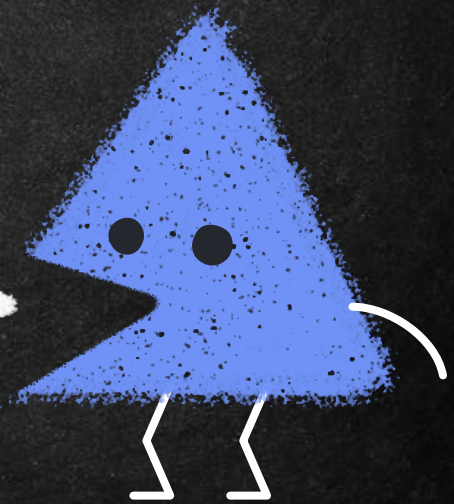
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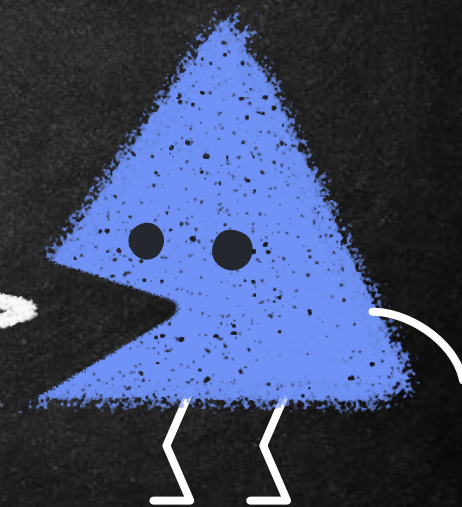
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 - The hard drive is connected to the read-only port of a firmware write block.
 - A target device to store an image or a copy is connected to the read-write port of the write block.



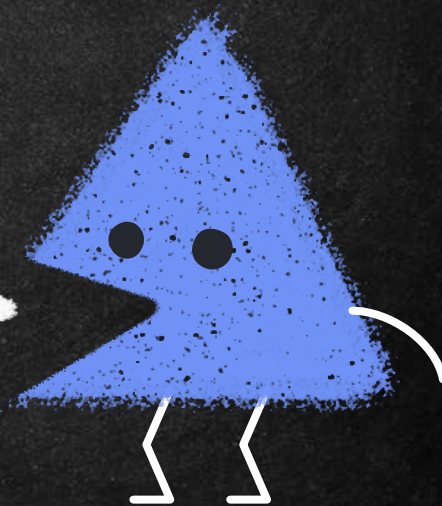
Data Acquisition

- The communications port of the write block is connected to a computer or a stand alone device, or is built in.
- The analyst then creates an image or a copy by writing to the target device.
- The original is maintained and documented in a chain-of-custody form.



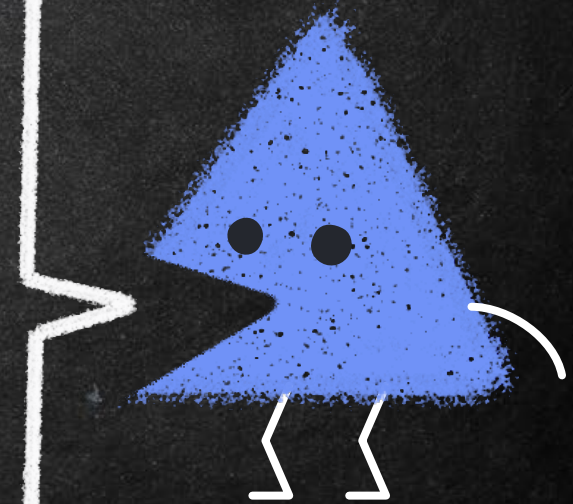
Data Acquisition

- The image or copy can then be examined (or additional copies made) on the analysis platform.
- This is typically done as a read-only operation as well, which allows for repeatability of any examination or experiments without modifying the data.



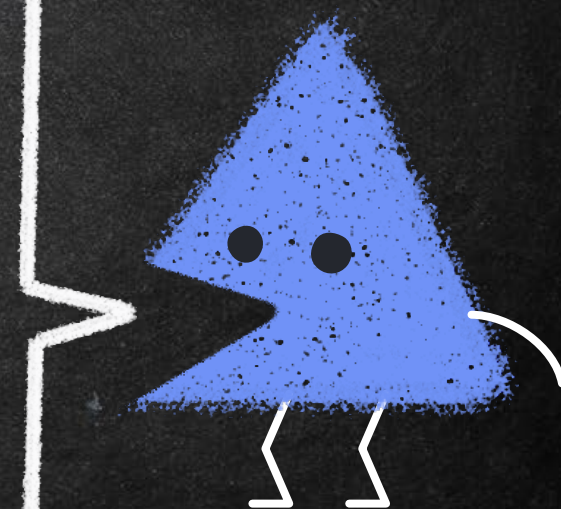
Chain of Custody

- A computer taken in as evidence makes many stops on its road to trial. It's collected, logged in at the lab, stored, checked out for analysis, checked back in for storage, and so on.
- Each of these stops must be noted, tracking each and every time the evidence item changes hands or locations.
- Without this detailed accounting, the evidence will be deemed untrustworthy and inadmissible.
- It's this detailed trail that makes up the chain of custody.



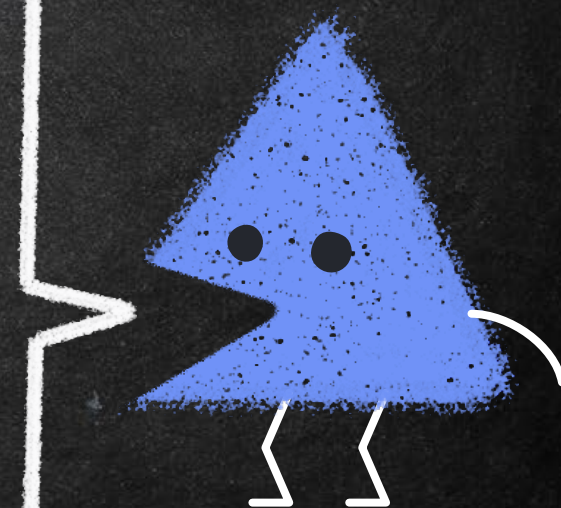
Hash Value

- hash is a unique value generated by a cryptographic hashing algorithm.
- Hash values (functions) are used in a variety of ways including cryptography and evidence integrity.
- Hash values are commonly referred to as a “digital fingerprint” or “digital DNA.”
- Any change to the hard drive, even by a single bit, will result in a radically different hash value. Therefore, any tampering or manipulation of the evidence is readily detectable.
- The most common hash functions used in digital forensics are Message Digest 5 (MD5), and Secure Hashing Algorithm (SHA) 1 and 2.



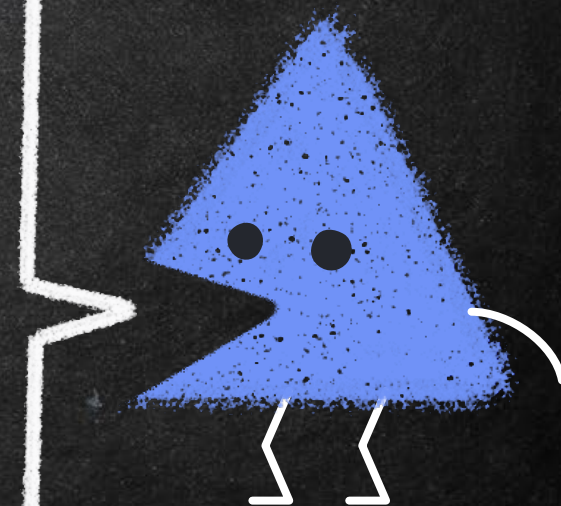
Use of Hash Value

- Hash values can be used throughout the digital forensic process.
- They can be used after the cloning process to verify that the clone is indeed an exact duplicate.
- They can also be used as an integrity check at any point it is needed.
- Examiners often have to exchange forensic images with the examiner on the opposing side. A hash value is sent along with the image so that it can be compared with the original.



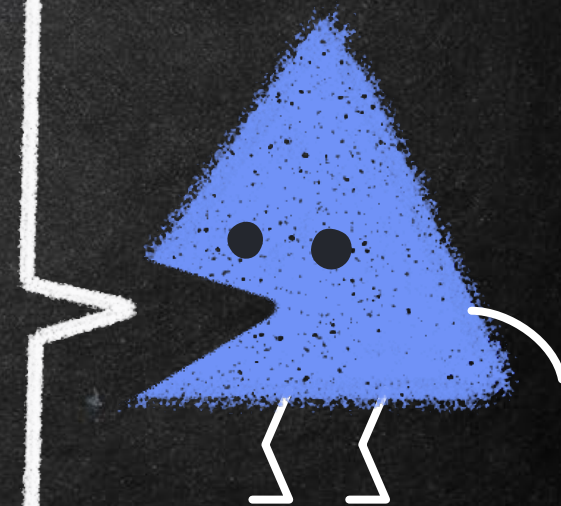
Use of Hash Value

- This comparison verifies that the image is a bit for bit copy of the original.
- The relevant hash values that were generated and recorded throughout the case should be kept and included with the final report. These digital fingerprints are crucial to demonstrating the integrity of the evidence and ultimately getting them before the jury.



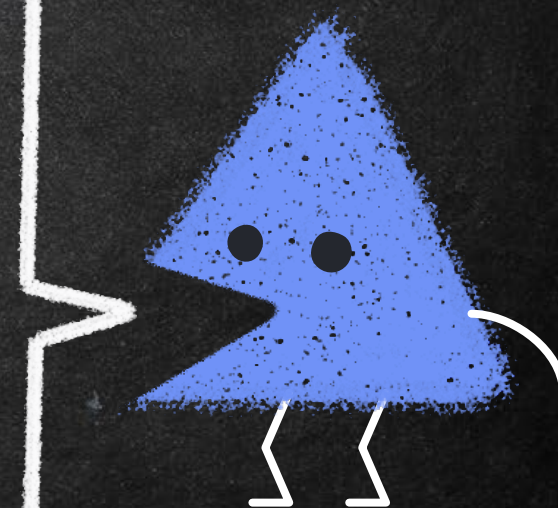
File Hashing

- File hashing is the act of attempting to uniquely identify a file.
- Different hash algorithms can be used for this purpose, but some are weaker than others and are more likely to have collisions.
- Once a file has been hashed, it can be compared with other file hashes.
- A common technique in incident response is to search for a file's hash value on Google and determine if the file's been seen before, and if so what its purpose is.



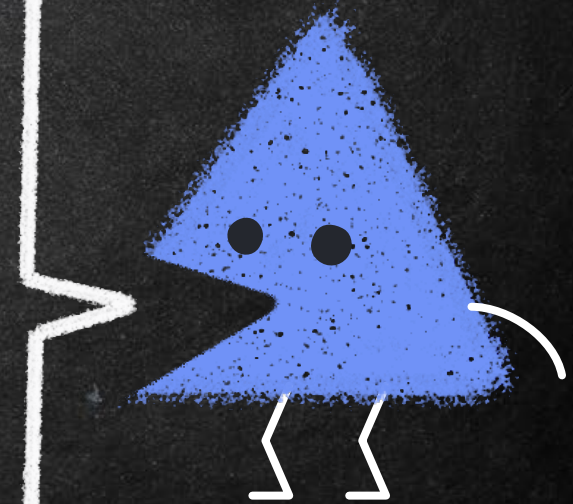
File Hashing

- Another technique used in contraband cases is to match a file's hash value with that of known contraband images to determine if the file had been seen before.
- In the case of trying to determine if a file is child sexually abusive material, identifying a file by hash value can help law enforcement prove the images are of a known victim.
- File hashing can also be used to whitelist files based on their signature. The white-listing process can result in data reduction in a case by removing legitimate files from a timeline. Doing so runs the risk of removing data from the analyst's view because attackers may use



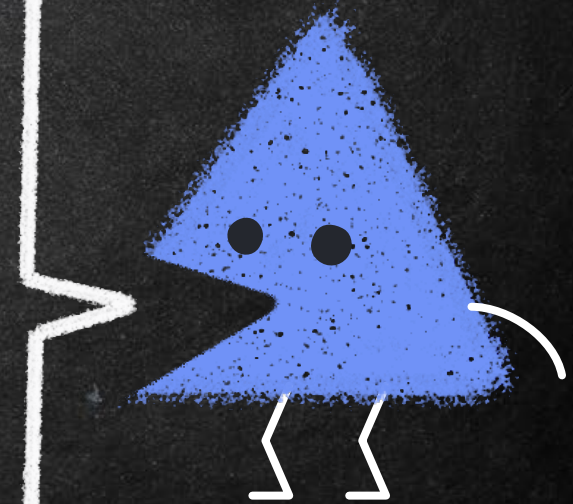
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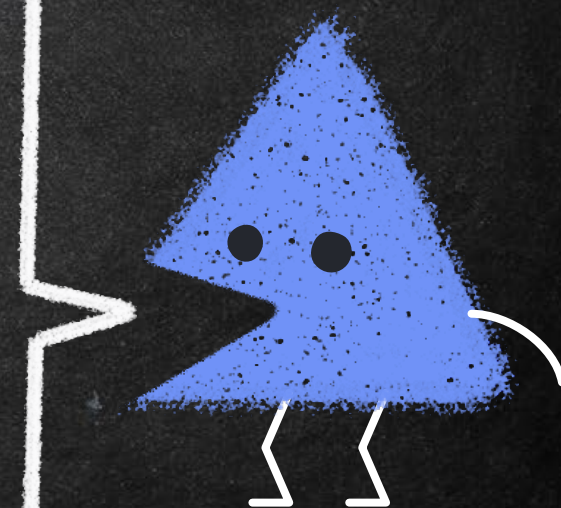
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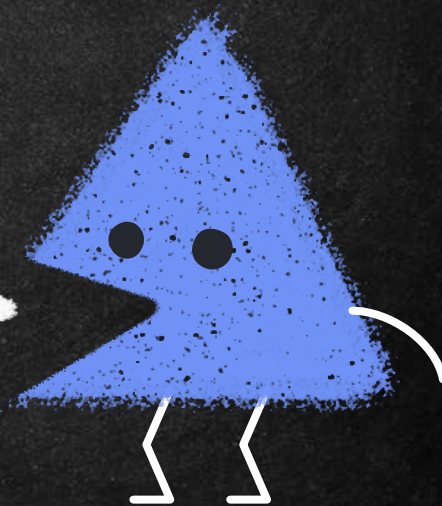
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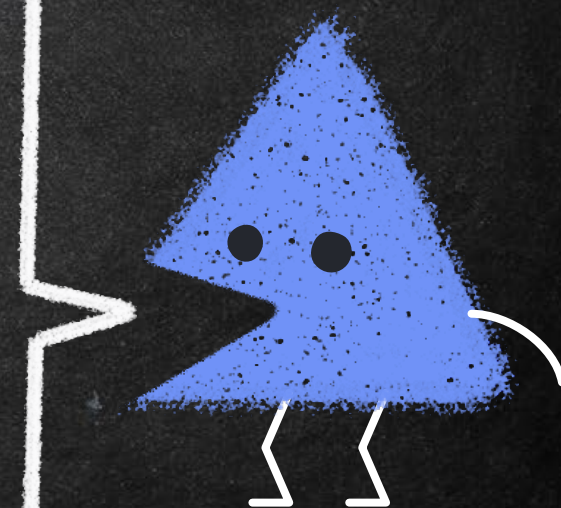
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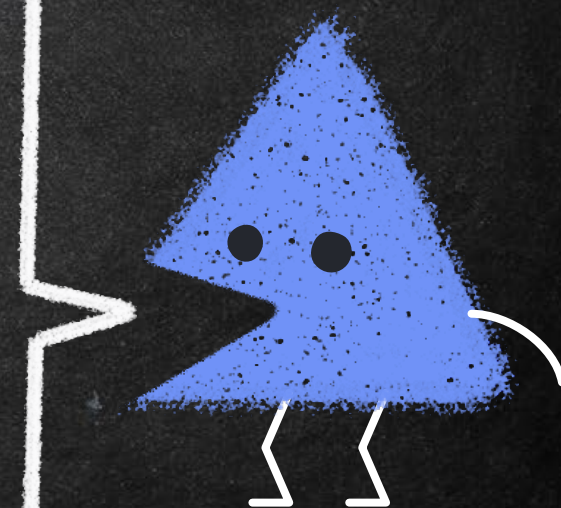
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References

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- Practical Forensic Imaging: Securing Digital Evidence with Linux Tools by Bruce Nikkel
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